

Choosing Linear, Quadratic, or Piecewise Models

Answer Key

Algebra 1 — Modeling and Functions

Quick Answers

- | | |
|---------|-----------------------------------|
| 1. 115 | 7. 8 |
| 2. 10 | 8. 10, 20 |
| 3. 28 | 9. $x = 8, y = 84$ |
| 4. 0, 4 | 10. $x = 1, y = 4, x = 5, y = 12$ |
| 5. 13 | 11. 89 |
| 6. 52 | 12. 20, 16 |
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Common wrong answers

1. *If they got 240: scaled the one-hour total by 6 instead of adding six equal increases to the start-up value*
 3. *If they got 168: used the first table step of 48 feet as a constant rate of change and extended it linearly*
 5. *If they got 19: charged the hourly rate for all five hours instead of only the hours past the first two*
 11. *If they got 29: read the first difference of 4 as a constant rate of change and extended the table linearly from the start value*
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1. Tutoring price table — cost of 6 hours.

Evidence first. The first differences are $40 - 25 = 15$, $55 - 40 = 15$, $70 - 55 = 15$ — constant, so the model is linear.

The value at $h = 0$ is the start-up charge, 25 dollars, and the constant increase is the rate, 15 dollars per hour:

$$C(h) = 25 + 15h$$

Evaluate at $h = 6$: $25 + 15(6) = 25 + 90 = 115$.

115

Common wrong answer 240: the one-hour total of 40 was multiplied by 6. That charges the 25 dollar start-up fee six times. Only the *rate* repeats; the intercept happens once.

2. Tutoring — how many hours cost 175 dollars?

Same linear model, now solved rather than evaluated:

$$25 + 15h = 175$$

$$15h = 150$$

$$h = 10$$

$$\boxed{h = 10}$$

Check: $25 + 15(10) = 175$ ✓

3. Ball height at $t = 3.5$ seconds.

Evidence: first differences 48, 16, -16 are not constant, but the second differences are $16 - 48 = -32$ and $-16 - 16 = -32$ — constant. Constant second differences mean a quadratic model, which is why $h(t) = -16t^2 + 64t$ is the right form.

Substitute $t = 3.5$:

$$h(3.5) = -16(3.5)^2 + 64(3.5) = -16(12.25) + 224 = -196 + 224 = 28$$

$$\boxed{28 \text{ ft}}$$

Common wrong answer 168 ft: the first step of 48 feet was used as a constant rate, giving 48×3.5 . A linear extrapolation from a quadratic table runs away from the truth quickly — here it puts the ball six times too high, and above its own maximum.

4. When is the ball at ground level?

Ground level means $h(t) = 0$. Factor rather than expand:

$$-16t^2 + 64t = 0$$

$$-16t(t - 4) = 0$$

The zero-product property gives $t = 0$ or $t = 4$.

$$\boxed{t = 0 \text{ or } t = 4}$$

Both roots are meaningful here: $t = 0$ is the launch and $t = 4$ is the landing. The vertex sits halfway between them at $t = 2$, which matches the table's maximum of 64 ft.

5. Parking garage — 5 hours.

The rule changes at a stated boundary, so the model is piecewise: a flat 4 dollars covers the first 2 hours, and only the hours *past* 2 are charged at 3 dollars each.

Hours past the boundary: $5 - 2 = 3$.

$$4 + 3(5 - 2) = 4 + 9 = 13$$

$$\boxed{13}$$

Common wrong answer 19: the hourly rate was applied to all five hours, $4 + 3(5)$. The flat fee already pays for the first two hours; charging them again is the classic piecewise slip.

6. Pool depth.

(a) First differences: $12 - 7 = 5$, $17 - 12 = 5$, $22 - 17 = 5$. Constant, so the model is linear with slope 5.

At $x = 1$ the depth is 7, so $7 = 5(1) + b$ and $b = 2$:

$$D(x) = 5x + 2$$

At $x = 10$: $5(10) + 2 = 52$.

52

(b) Model answer: the table's *first* differences are constant — the depth rises by exactly 5 cm every minute. A quadratic model would show first differences that change from step to step, with the *second* differences constant instead. Since the first differences never change, no x^2 term is needed, and a straight line fits every point in the table.

Manual review: part (b) is a written justification, read by a teacher; part (a) is machine-checked. Credit an answer that names constant first differences and contrasts them with what a quadratic table would look like.

7. Parking garage — a 22 dollar ticket.

Same piecewise rule, solved for the input. A 22 dollar charge is well past the flat-fee region, so the second piece applies:

$$4 + 3(t - 2) = 22$$

$$3(t - 2) = 18$$

$$t - 2 = 6$$

$$t = 8$$

$t = 8$

Check the piece: $8 > 2$, so the second piece really is the one in force. On a piecewise problem the answer is only valid if it lands in the interval whose formula produced it.

8. Hoodie revenue of 400 dollars.

Revenue is price times quantity, and the quantity itself depends on the price — a product of two linear expressions, which is exactly what makes the model quadratic.

$$p(60 - 2p) = 400$$

$$60p - 2p^2 = 400$$

$$-2p^2 + 60p - 400 = 0$$

$$p^2 - 30p + 200 = 0$$

$$(p - 10)(p - 20) = 0$$

$p = 10$ or $p = 20$

Two prices give the same revenue — a low price selling many hoodies (10×40) and a high price selling few (20×20). A linear model could never produce two answers, which is itself evidence the model is quadratic.

9. Two gyms — when do the plans cost the same?

Both stories name a fixed fee plus a fixed monthly rate, so both models are linear and the question is where they meet:

$$\begin{cases} y = 20 + 8x \\ y = 44 + 5x \end{cases}$$

Substitute the first into the second:

$$\begin{aligned} 20 + 8x &= 44 + 5x \\ 3x &= 24 \\ x &= 8 \end{aligned}$$

Then $y = 20 + 8(8) = 84$, and the check on the other model gives $44 + 5(8) = 84$ ✓

$$\boxed{x = 8, y = 84}$$

Reading the answer: before 8 months Gym A is cheaper (its joining fee is smaller); after 8 months Gym B is cheaper (its rate is smaller). The intersection is where the advice changes.

10. Where a quadratic model and a linear model agree.

Agreement means equal outputs at the same input, so set the two models equal and solve:

$$\begin{aligned} x^2 - 4x + 7 &= 2x + 2 \\ x^2 - 6x + 5 &= 0 \\ (x - 1)(x - 5) &= 0 \end{aligned}$$

So $x = 1$ or $x = 5$. Substitute each into the *linear* model, which is the easier of the two:

$$x = 1: y = 2(1) + 2 = 4. \quad x = 5: y = 2(5) + 2 = 12.$$

$$\boxed{(1, 4) \text{ and } (5, 12)}$$

Check in the quadratic: $1 - 4 + 7 = 4$ ✓ and $25 - 20 + 7 = 12$ ✓. A line and a parabola can meet twice, once, or not at all — getting two points here is normal, not an error.

11. Population — finding Rae's mistake.

What Rae's evidence actually shows: the first differences are $9 - 5 = 4$, $17 - 9 = 8$, $29 - 17 = 12$. They are *not* constant, so the growth is not at a fixed rate and no linear model fits. The second differences are $8 - 4 = 4$ and $12 - 8 = 4$ — constant, which is the signature of a quadratic model. Rae used the very first difference as though it held forever.

With the correct model $P(t) = 2t^2 + 2t + 5$:

$$P(6) = 2(36) + 2(6) + 5 = 72 + 12 + 5 = 89$$

$$\boxed{89}$$

Rae's answer, 29, is exactly $5 + 4 \times 6$ — the linear extrapolation. It also happens to be the year-3 population, so a student who sanity-checks against the table should notice the model has stopped growing.

12. Synthesis — the rideshare fare.

The rule changes at 6 miles, so the model is piecewise:

$$F(m) = \begin{cases} 3 + 2m, & 0 \leq m \leq 6 \\ 3 + 12 + 1.25(m - 6), & m > 6 \end{cases}$$

(a) A 10 mile ride is past the boundary, so use the second piece. The first 6 miles cost $2 \times 6 = 12$ dollars, and the remaining $10 - 6 = 4$ miles cost 1.25 each:

$$3 + 12 + 1.25(10 - 6) = 3 + 12 + 5 = 20$$

20

(b) A 27.50 dollar fare is larger than the 15 dollar fare at the boundary, so the second piece is again in force:

$$3 + 12 + 1.25(m - 6) = 27.5$$

$$1.25(m - 6) = 12.5$$

$$m - 6 = 10$$

$$m = 16$$

$m = 16$

(c) Model answer: a linear function has one constant rate of change everywhere, but this fare has two — 2 dollars per mile up to 6 miles and 1.25 dollars per mile after that. No single slope can be both. The graph is two connected straight segments: a steeper one from $(0, 3)$ to $(6, 15)$, then a shallower one continuing from $(6, 15)$ onward. The two segments meet without a gap, so the fare is continuous, but the bend at 6 miles is a corner, and that corner is exactly what a single line cannot have.

Manual review: part (c) is written reasoning, read by a teacher; parts (a) and (b) are machine-checked. Credit an answer that names the two different rates and describes a bend at the 6 mile boundary.